

National Higher School of Mathematics

Preparatory Cycle
First Year

Academic Year : 2023/2024
Course : Algebra 1

Worksheet 2 : Sets and Maps

Exercise 1.

Which of the following statements are true and which are false ?

1. $\{a, \emptyset\} \subset \{a, \{a, \emptyset\}\}.$
2. $1 \in \mathcal{P}(\{1, 2\}).$
3. $\emptyset \cap \{\emptyset\} = \emptyset.$
4. If $A := \{a, \{a\}\}$, then $\{\{a\}\} \in \mathcal{P}(A).$
5. $\{x \in \mathbb{N} \mid x^3 = x^2 + 5x + 3\} = \{-1, 3\}.$
6. $\emptyset \in A$, for any set A .

Exercise 2.

1. Consider the following subsets of $\mathbb{Z}$.

$$\begin{aligned} A &:= \{n \in \mathbb{Z} \mid n = 2m \text{ for some } m \in \mathbb{Z}\} & B &:= \{n \in \mathbb{Z} \mid 3 \text{ divides } n\} \\ C &:= \{n \in \mathbb{Z} \mid n^2 \text{ is odd}\} & D &:= \{n \in \mathbb{Z} \mid 0 \leq n \leq 10\}. \end{aligned}$$

Find each of the following sets

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|-----------------|----------------------|-------------------------------|
| (a) $A \cup C.$ | (c) $A \cap B.$ | (e) $C \setminus B.$ |
| (b) $A \cap C.$ | (d) $D \setminus A.$ | (f) $D \cap (A \setminus B).$ |

2. In each of the following cases, suppose we are given sets A_k for each $k \in \mathbb{N}$. Find $\bigcap_{k \in \mathbb{N}} A_k$ and $\bigcup_{k \in \mathbb{N}} A_k$.

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|--|---|
| (a) $A_k = \{0, 1, 2, 3, \dots, 2k\}.$ | (d) $A_k = \left[\frac{1}{k}, \frac{3k+2}{k}\right) \cup \{5+k\}.$ |
| (b) $A_k = \{k-1, k, k+1\}.$ | (e) $A_k = \left[-1, 3 + \frac{1}{k}\right] \cup \left(2, \frac{3k-1}{k}\right].$ |
| (c) $A_k = \left[0, \frac{k-3}{k+2}\right].$ | |

Exercise 3.

Let S be a set and let A, B, C be subsets of S .

1. If we have $C \subset A \cup B$, can we say that $C \subset A$ or $C \subset B$?
2. What can we say about the sets A and B if $A \cap B = A \cup B$. Justify your answer.
3. Show that
 - (a) $A \cup B = B \cap C \implies A \subset B \subset C.$
 - (b) $(A \cap B = A \cap C \wedge A \cup B = A \cup C) \implies B = C.$
 - (c) $A \subset C \iff A \cup (B \cap C) = (A \cup B) \cap C.$
 - (d) $(A \cap \overline{B} = A \cap \overline{C}) \iff (A \cap B = A \cap C).$
 - (e) $(A \cap B \subset A \cap C) \wedge (A \cup B \subset A \cup C) \implies B \subset C.$
4. (a) Simplify the following expressions :

- i. $(\overline{A} \cap B) \cup (\overline{A} \cap \overline{B}) \cup (A \cup B)$.
 ii. $(\overline{A} \cap B) \cup (\overline{A} \cap \overline{B}) \cup (A \cap B)$.
 iii. $(A \cup B) \setminus (A \cup C)$.

(b) Deduce that $(A \cup B) \Delta (A \cup C) = \overline{A} \cap (B \Delta C)$.

Exercise 4.

Let A, B, C be subsets of a set S . Show that :

1. $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$.
2. $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$.
3. $A \cap (B \setminus C) = (A \cap B) \setminus (A \cap C)$.
4. $A \setminus B = A \iff B \setminus A = B$.
5. $(A \setminus C) \setminus (B \setminus C) = (A \setminus B) \setminus C$.
6. $(A \setminus B) \setminus C = A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.
7. $A \Delta B = (A \cup B) \setminus (A \cap B)$.
8. $(A \cap B) \Delta (A \cap C) = A \cap (B \Delta C)$.
9. $\overline{A} \Delta \overline{B} = A \Delta B$.
10. $A \Delta B = A \cap B \iff A = B = \emptyset$.

Exercise 5.

Let S be a set and $\emptyset \neq E \subset \mathcal{P}(S)$. Show that the following proprieties are equivalent.

1. $\forall (A, B) \in E^2, A \Delta B \in E$ and $A \cap B \in E$.
2. $\forall (A, B) \in E^2, A \Delta B \in E$ and $A \cup B \in E$.
3. $\forall (A, B) \in E^2, A \cup B \in E$ and $A \setminus B \in E$.

Exercise 6.

1. Let $A := \{0, 1\}$, $B := \{-2, 1\}$ and $C := \{3, 5, 7\}$. Determine the following sets

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|-------------------------------|---------------------------------|--|
| (a) $A \times B \times C$. | (c) $(A \times B) \times C$. | (e) $\mathcal{P}(A \times B)$. |
| (b) $A \times (B \times C)$. | (d) $A \times \mathcal{P}(B)$. | (f) $\mathcal{P}(A) \times \mathcal{P}(B)$. |

2. Let $D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$. Show that D can not be written as a Cartesian product $A \times B$, where $A, B \subset \mathbb{R}$.

Exercise 7.

Let A, B, C, D, E and F be non-empty sets. Prove the following identities for Cartesian products :

1. $(A \subset C) \wedge (B \subset D) \iff A \times B \subset C \times D$.
2. $(A \times B) \cup (A \times C) = A \times (B \cup C)$.
3. $(A \times B) \cap (A \times C) = A \times (B \cap C)$.
4. $(A \times B) \setminus (A \times C) = A \times (B \setminus C)$.
5. $(E \times H) \cap (F \times G) = (E \cap F) \times (H \cap G)$.
6. If $A \neq B$ and $A \times C = B \times C$, then $C = \emptyset$.

Exercise 8.

For any non-empty subset A of a set S , the characteristic function of A is the map f from S to $\{0, 1\}$ defined as follows

$$f(x) := \begin{cases} 1, & \text{if } x \in A, \\ 0, & \text{otherwise.} \end{cases}$$

Let A and B be non-empty two subsets of a set S . Denote by f and g the characteristic functions of A and B respectively. Show that the maps $1 - f$, $f \times g$, and $f + g - f \times g$ are characteristic functions.

Exercise 9.

Are the following maps injective ? surjective ? bijective ? Determine the inverse map when possible.

$$\begin{array}{lll} f_1: \mathbb{R} \longrightarrow \mathbb{R} & f_2: \mathbb{R} \longrightarrow \mathbb{R} & f_3: \mathbb{R} \longrightarrow \mathbb{R}_+^* \\ x \longmapsto f_1(x) = x - E(x); & x \longmapsto f_2(x) = E(2x); & x \longmapsto f_3(x) = \frac{e^x + 2}{e^{-x}}; \end{array}$$

$$f_4: \mathbb{R} \longrightarrow]-1, 1[\\ x \longmapsto f_4(x) = \frac{x}{1 + |x|}; \quad f_5: \mathbb{R}^2 \longrightarrow \mathbb{R} \\ (x, y) \longmapsto f_5(x) = x + y; \quad f_6: \mathbb{R}^2 \longrightarrow \mathbb{R}^2 \\ (x, y) \longmapsto f_6(x, y) = (x + y, x - y);$$

$$f_7: \mathbb{R}^2 \longrightarrow \mathbb{R}^2 \\ (x, y) \longmapsto f_7(x, y) = (x + y, xy); \quad f_8: \mathbb{N}^2 \longrightarrow \mathbb{N} \\ (n, p) \longmapsto f_8(n, p) = 2^n(2p + 1).$$

Exercise 10.

Let f be the map defined as

$$f: \mathbb{R} \longrightarrow \mathbb{R} \\ x \longmapsto \frac{x}{\sqrt{1 + x^2}}.$$

1. Show that f is injective. Is it surjective?
2. Determine the sets $f^{-1}([1, \infty[)$, $f([1, 2])$ and $f([-2, 2])$.
3. Show that $f|^{]-1, 1[}$ is bijective and determine its inverse map.
4. Let g be the restriction of $f|^{]-1, 1[}$ to $]-1, 1[$ and for any positive integer n , we denote by $g^{(n)}$ the map $\underbrace{g \circ g \circ \dots \circ g}_{n \text{ times}}$. Determine $g^{(n)}(x)$ for $n \geq 2$.

Exercise 11.

Let A and B be two subsets of a set S , and f be the following map :

$$f: \mathcal{P}(S) \longrightarrow \mathcal{P}(A) \times \mathcal{P}(B) \\ X \longmapsto (A \cap X, B \cap X).$$

1. Prove that f is injective if and only if $A \cup B = S$.
2. Prove that f is surjective if and only if $A \cap B = \emptyset$.
3. Give the expression of f^{-1} when it exists.

Exercise 12.

Let f be the map defined as follows :

$$f: \mathbb{N} \setminus \{1\} \longrightarrow \mathbb{R} \\ n \longmapsto \frac{n^2 - n + 2}{n - 1}.$$

1. Calculate $f(A)$, $f^{-1}(B)$, $f(C)$, and $f^{-1}(\mathbb{Z})$, where

$$A := \left\{ \frac{1}{3}, 4 \right\}, \quad B :=]-\infty, -1], \quad \text{and} \quad C := \{n \in \mathbb{N} \setminus \{1\} \mid |n^3 - n + 1| \leq 1\}.$$

2. Let

$$H := \left\{ x \in \mathbb{Q} \mid \frac{1}{x - E(x)} \in \mathbb{N} \right\}.$$

Show that $f^{-1}(H) = \{2n + 1 \mid n \in \mathbb{N}, n \geq 2\}$.

Exercise 13 (Cantor's theorem).

Let S be a set.

1. Determine an injective map from S to $\mathcal{P}(S)$.
2. Prove that there is no bijective map from S to $\mathcal{P}(S)$. (Hint : consider a surjective map $f : S \rightarrow \mathcal{P}(S)$ and determine the inverse image of the set $\{x \in S \mid x \notin f(x)\}$).

Exercise 14.

Let X, Y and Z be three sets and let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two maps. Show that

1. If $g \circ f$ is injective then f is injective.
2. If $g \circ f$ is injective and f is surjective then g is injective.
3. If $g \circ f$ is surjective then g is surjective.
4. If $g \circ f$ is surjective and g is injective then f is surjective.
5. For any subset A of X , $A \subset f^{-1}(f(A))$.
6. f is injective if and only if for any subset A of X , $f^{-1}(f(A)) \subset A$.
7. For any subset B of Y , $f(f^{-1}(B)) \subset B$.
8. f is surjective if and only if for any subset B of Y , $B \subset f(f^{-1}(B))$.

Exercise 15.

Let X, Y be two sets and let $f : X \rightarrow Y$ be a map. Consider the subsets $A, A_1, A_2 \subset X$ and $B, B_1, B_2 \subset Y$.

1. (a) Prove that $A_1 \subset A_2 \implies f(A_1) \subset f(A_2)$.
(b) Find an example showing that the implication $f(A_1) \subset f(A_2) \implies A_1 \subset A_2$ is not always true.
2. Prove that $f(A_1 \cup A_2) = f(A_1) \cup f(A_2)$.
3. (a) Prove that $f(A_1 \cap A_2) \subset f(A_1) \cap f(A_2)$.
(b) Find an example showing that the equality is not always true.
(c) Prove that f is injective if and only if $f(A_1 \cap A_2) = f(A_1) \cap f(A_2)$.
4. Find a counter-example to show that the following statements are false :

$$(a) f(\overline{A}) \subset \overline{f(A)}. \quad (b) \overline{f(A)} \subset f(\overline{A}).$$

5. Show that f is bijective if and only if $f(\overline{A}) = \overline{f(A)}$, for any $A \subset X$.
6. Prove that

$$(a) f^{-1}(B_1 \cup B_2) = f^{-1}(B_1) \cup f^{-1}(B_2). \quad (c) f^{-1}(\overline{B}) = \overline{f^{-1}(B)}.$$

$$(b) f^{-1}(B_1 \cap B_2) = f^{-1}(B_1) \cap f^{-1}(B_2).$$

Exercise 16.

Let $f : X \rightarrow Y$ be a map. Show that

1. The map f is injective if and only if there exists a map $h_1 : Y \rightarrow X$, such that $h_1 \circ f = \text{Id}_X$.
2. The map f is surjective if and only if there exists a map $h_2 : Y \rightarrow X$, such that $f \circ h_2 = \text{Id}_Y$.

Exercise 17 (Some countable sets).

The purpose of this exercise is to show that the sets $\mathbb{N}_0^2$, $\mathbb{Z}$, $\mathbb{N}$, $\mathbb{Z} \times \mathbb{N}$ and $\mathbb{Q}$ are countable.

1. Let $\varphi : \mathbb{N}_0^2 \rightarrow \mathbb{N}$ be the map defined as follows :

$$\forall (p, q) \in \mathbb{N}_0^2, \varphi(p, q) = (2q + 1)2^p.$$

- (a) Show that φ is bijective.
- (b) Deduce that $\mathbb{N}_0^2$ is countable.
2. Let $f : \mathbb{Z} \longrightarrow \mathbb{N}_0$ and $g : \mathbb{N} \longrightarrow \mathbb{N}_0$ be respectively the maps defined by :

$$\forall n \in \mathbb{Z}, f(n) := \begin{cases} 2n, & \text{if } n \geq 0, \\ -2n - 1, & \text{if } n < 0, \end{cases}$$

and $\forall m \in \mathbb{N}, g(m) = m - 1$.

- (a) Show that the maps f and g are bijective.
- (b) Deduce that $\mathbb{Z}$, $\mathbb{N}$ and $\mathbb{Z} \times \mathbb{N}$ are countable sets.
3. (a) Let S be a set and let $f : S \longrightarrow \mathbb{N}_0$ be a map. Show that if f is injective then S is countable.
- (b) Find an injective map from $\mathbb{Q}$ into $\mathbb{Z} \times \mathbb{N}$. Conclude.